

Home Problem 1

Stochastic Optimization Algorithms
Daniel Peña Fonseca
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Problem 1.1

The function given in this problem is:

$$f(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2 \quad (1)$$

With the constraint:

$$g(x_1, x_2) = x_1^2 + x_2^2 - 1 \leq 0 \quad (2)$$

1. The penalty function, $f_p(x_1, x_2)$ is defined as:

$$f_p(x_1, x_2) = f(x_1, x_2) + P(x_1, x_2) \quad (3)$$

Where $P(x_1, x_2)$ is the penalty term.

To define the penalty term $P(x_1, x_2)$, it is important to check whether the constraint presents an equality or an inequality. In this case, $g(x_1, x_2) \leq 0$, thus we want to penalize $g(x_1, x_2) > 0$:

$$\begin{cases} \text{if } g(x_1, x_2) > 0 \rightarrow P(x_1, x_2) = \mu[g(x_1, x_2)]^2 \\ \text{if } g(x_1, x_2) \leq 0 \rightarrow P(x_1, x_2) = 0 \end{cases} \quad (4)$$

Where μ is a parameter.

Therefore, the penalty function is defined as:

$$\begin{cases} \text{if } g(x_1, x_2) > 0 \rightarrow f_p(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2 + \mu[x_1^2 + x_2^2 - 1]^2 \\ \text{if } g(x_1, x_2) \leq 0 \rightarrow f_p(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2 \end{cases} \quad (5)$$

2. The gradient $\nabla f_p(\mathbf{x}; \mu)$ is defined as:

$$\nabla f_p(\mathbf{x}; \mu) = \left(\frac{\partial f_p}{\partial x_1}, \frac{\partial f_p}{\partial x_2} \right) \quad (6)$$

$$\frac{\partial f_p}{\partial x_1} \equiv \partial_{x_1} f_p = \begin{cases} \text{if } g(x_1, x_2) > 0 \rightarrow \partial_{x_1} f_p = 2(x_1 - 1) + \mu \cdot 2[x_1^2 + x_2^2 - 1] \cdot 2x_1 \\ \text{if } g(x_1, x_2) \leq 0 \rightarrow \partial_{x_1} f_p = 2(x_1 - 1) \end{cases} \quad (7)$$

$$\frac{\partial f_p}{\partial x_2} \equiv \partial_{x_2} f_p = \begin{cases} \text{if } g(x_1, x_2) > 0 \rightarrow \partial_{x_2} f_p = 4(x_2 - 2) + \mu \cdot 2[x_1^2 + x_2^2 - 1] \cdot 2x_2 \\ \text{if } g(x_1, x_2) \leq 0 \rightarrow \partial_{x_2} f_p = 4(x_2 - 2) \end{cases} \quad (8)$$

3. Since we are interested in the minimum value, we have to search for the stationary points; i.e., those satisfying:

$$\nabla f_p(\mathbf{x}; \mu) = 0 \quad (9)$$

To search for the unconstrained minimum ($\mu = 0$), we can use the previous expressions for $g(x_1, x_2) \leq 0$, since they are equivalent to having $\mu = 0$. In this case:

$$\nabla f_p(\mathbf{x}; \mu)|_{\mathbf{x}^*} = \left(\frac{\partial f_p}{\partial x_1}, \frac{\partial f_p}{\partial x_2} \right) \Big|_{\mathbf{x}^*} = [2(x_1 - 1), 4(x_2 - 2)]|_{\mathbf{x}^*} = 0 \quad (10)$$

From this equation we extract the stationary point, $\mathbf{x}^*$:

$$\mathbf{x}^* = (x_1^*, x_2^*) = (1, 2) \quad (11)$$

We now do the Hessian matrix to analyze the nature of the stationary point that has been found.

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1}(2(x_1 - 1)) & \frac{\partial}{\partial x_1}(4(x_2 - 2)) \\ \frac{\partial}{\partial x_2}(2(x_1 - 1)) & \frac{\partial}{\partial x_2}(4(x_2 - 2)) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \quad (12)$$

Since the Hessian matrix is positive semi-definite, the function $f(x_1, x_2)$ is convex. And therefore, the stationary point found $\mathbf{x}^* = (1, 2)$ is a minimum.

4. The Python program is presented in the folder *Problem_1.1*

5. Using the Python program presented in the folder *Problem_1.1*, we detect convergence of the values (x_1^*, x_2^*) that minimize the function $f_p(x_1, x_2)$, as presented in Figure 1. The most representative values are presented in the Table 1 (see *Appendix A* for the complete table with all the values presented in Figure 1).

μ	x_1^*	x_2^*	$f_p(x_1^*, x_2^*)$	$g(x_1^*, x_2^*)$
10^{-3}	0.9922	1.9921	0.0002	3.9529
10^{-2}	0.9329	1.9306	0.0142	3.5973
10^{-1}	0.6963	1.6419	0.3487	2.1808
1	0.4338	1.2102	1.5683	0.6527
10	0.3314	0.9955	2.4650	0.1009
100	0.3137	0.9553	2.6540	0.0109
1000	0.3118	0.9507	2.6756	0.0011

Table 1: For each value of μ , the gradient descent method was performed to find the values (x_1^*, x_2^*) for which the function $f(x_1, x_2)$ takes its minimum value. The step length used for the gradient descent was 10^{-4} .

The reason for this convergence, where a larger μ leads to a better result (closer to the real minimum value given the constraint), can be observed by checking the form of the penalty function:

$$\begin{cases} \text{if } g(x_1, x_2) > 0 \rightarrow f_p(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2 + \mu[x_1^2 + x_2^2 - 1]^2 \\ \text{if } g(x_1, x_2) \leq 0 \rightarrow f_p(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2 \end{cases} \quad (13)$$

In this expression we can see how, the greater μ , the more penalized it will be to not fulfill the constraint marked by $g(x_1, x_2)$, leading to smaller values of $g(x_1^*, x_2^*)$. Consequently, the smaller μ , the less importance the penalty will have; until, for $\mu \rightarrow 0$, we obtain $f_p(x_1, x_2) \rightarrow f(x_1, x_2)$, recovering the result $(x_1^* = 1, x_2^* = 2)$ that we obtained analytically for the unconstrained problem. This behavior can also be observed in Figure 1.

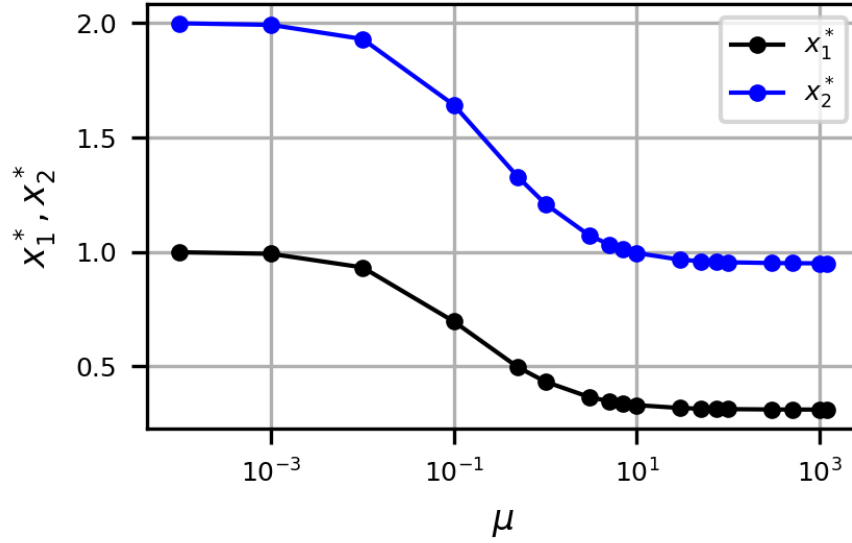


Figure 1: For each value of μ , the gradient descent method was used to find the values (x_1^*, x_2^*) for which the function $f_p(x_1, x_2)$ takes its minimum value. These values seem to converge towards $(x_1 \approx 0.3118, x_2 \approx 0.9506)$ for higher values of μ , while converging towards $(x_1 = 1, x_2 = 2)$ for $\mu \rightarrow 0$. The step length used for the gradient descent was 10^{-4} .

Problem 1.2

Objective: Determine the values $[(x_1^*, x_2^*), f(x_1^*, x_2^*)]$ where the maximum and minimum values of $f(x_1, x_2)$ appear. For this problem, $f(x_1, x_2) = 4x_1^2 + 2x_2^3$ over the closed set S defined by $x_1^2 + x_2^2 \leq 4$.

When analyzing the boundary ∂S , the Lagrange multiplier method should be used.

1. Interior of S

To analyze the minimum values, we first have to extract the stationary points, $\mathbf{x}^*$, which satisfy $\nabla f|_{\mathbf{x}^*} = 0$:

$$\nabla f|_{\mathbf{x}^*} = \left(\frac{\partial f_p}{\partial x_1}, \frac{\partial f_p}{\partial x_2} \right) \Big|_{\mathbf{x}^*} = (8x_1, 6x_2^2) \Big|_{x_1^*, x_2^*} = (0, 0) \Rightarrow \mathbf{x}^* \equiv (x_1^*, x_2^*) = (0, 0) \quad (14)$$

As we did in the previous exercise, we now do the Hessian matrix to analyze the nature of the stationary point that has been found.

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1}(8x_1) & \frac{\partial}{\partial x_1}(6x_2^2) \\ \frac{\partial}{\partial x_2}(8x_1) & \frac{\partial}{\partial x_2}(6x_2^2) \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 12x_2 \end{pmatrix} \quad (15)$$

Now we evaluate the Hessian matrix in the stationary point $\mathbf{x}^*$:

$$H|_{\mathbf{x}^*} = \begin{pmatrix} 8 & 0 \\ 0 & 0 \end{pmatrix} \quad (16)$$

Since the Hessian matrix evaluated in the stationary point is positive semi-definite, i.e., one eigenvalue is $\lambda_1 > 0$ but the other one is $\lambda_2 = 0$, we cannot define this stationary point as a minimum, but we can guarantee that it is not a maximum.

This point is inside the set S , since it satisfies $x_1^2 + x_2^2 \leq 4$:

$$(x_1^*)^2 + (x_2^*)^2 \leq 4 \rightarrow 0 + 0 < 4 \blacksquare \quad (17)$$

Therefore:

$$\text{Stationary point in: } (x_1 = 0, x_2 = 0), \text{ with } f(0, 0) = 0 \quad (18)$$

We will define this stationary point as $P_0 = (x_1 = 0, x_2 = 0)$. This notation will facilitate comparison with the points calculated when analyzing the boundary of the set, ∂S .

2. Boundary ∂S

Using the Lagrange method, we will calculate the stationary points in the boundary ∂S . Now we define:

$$L(x_1, x_2; \lambda) = 4x_1^2 + 2x_2^3 + \lambda(x_1^2 + x_2^2 - 4) \quad (19)$$

Setting the derivatives of L with respect to x_1 , x_2 and λ to 0, we can calculate the stationary points.

$$\frac{\partial L}{\partial x_1} = 8x_1 + \lambda \cdot 2x_1 = 0 \quad (20)$$

$$\frac{\partial L}{\partial x_2} = 6x_2^2 + \lambda \cdot 2x_2 = 0 \quad (21)$$

$$\frac{\partial L}{\partial \lambda} = x_1^2 + x_2^2 - 4 = 0 \quad (22)$$

Given these equations, we cannot have $x_1 = x_2 = 0$, since eq. (22) would not be satisfied. But we can have $(x_1 = 0, x_2 \neq 0)$, $(x_1 \neq 0, x_2 = 0)$, and $(x_1 \neq 0, x_2 \neq 0)$.

We start with $x_1 = 0, x_2 \neq 0$:

$$x_1 = 0, x_2 \neq 0 \Rightarrow \text{eq.(21)} \rightarrow \lambda = -\frac{6x_2^2}{2x_2} = -3x_2; \text{eq.(22)} \rightarrow x_2 = \pm\sqrt{4} = \pm 2 \Rightarrow \lambda = \mp 6 \quad (23)$$

With these results, we define $P_1 = (x_1 = 0, x_2 = 2)$ and $P_2 = (x_1 = 0, x_2 = -2)$ as stationary points.

For the case of $x_1 \neq 0, x_2 = 0$:

$$x_1 \neq 0, x_2 = 0 \Rightarrow \text{eq.(20)} \rightarrow \lambda = \frac{-8x_1}{2x_1} = -4; \text{eq.(22)} \rightarrow x_1 = \pm\sqrt{4} = \pm 2 \quad (24)$$

We now define $P_3 = (x_1 = 2, x_2 = 0)$ and $P_4 = (x_1 = -2, x_2 = 0)$ as other stationary points.

Lastly, we study the case $x_1 \neq 0, x_2 \neq 0$:

$$x_1 \neq 0, x_2 \neq 0 \Rightarrow \text{eq.(20)} \rightarrow \lambda = \frac{-8x_1}{2x_1} = -4; \text{eq.(21)} \rightarrow \lambda = \frac{-6x_2^2}{2x_2} = -3x_2 \Rightarrow x_2 = 4/3 \quad (25)$$

Now using eq. (22):

$$x_1^2 + (4/3)^2 - 4 = 0 \Rightarrow x_1^2 + \frac{16}{9} - \frac{36}{9} = 0 \rightarrow x_1^2 = \frac{20}{9} \Rightarrow x_1 = \pm \frac{\sqrt{20}}{3} \quad (26)$$

From where we extract two more stationary points: $P_5 = (x_1 = \sqrt{20}/3, x_2 = 4/3)$ and $P_6 = (x_1 = -\sqrt{20}/3, x_2 = 4/3)$.

Now we analyze the values taken by the function on the stationary points in order to determine which one gives the maximum value of f , and which one returns the minimum value:

$$P_0 = (0, 0) \rightarrow f(P_0) = 0 \quad (27)$$

$$P_1 = (0, 2) \rightarrow f(P_1) = 2 \cdot (2)^3 = 16 \quad (28)$$

$$P_2 = (0, -2) \rightarrow f(P_2) = 2 \cdot (-2)^3 = -16 \quad (29)$$

$$P_3 = (2, 0) \rightarrow f(P_3) = 4 \cdot (2)^2 = 16 \quad (30)$$

$$P_4 = (-2, 0) \rightarrow f(P_4) = 4 \cdot (-2)^2 = 16 \quad (31)$$

$$P_5 = (\sqrt{20}/3, 4/3) \rightarrow f(P_5) = 4 \cdot (\sqrt{20}/3)^2 + 2 \cdot (4/3)^3 \approx 13.63 \quad (32)$$

$$P_6 = (-\sqrt{20}/3, 4/3) \rightarrow f(P_6) = 4 \cdot (-\sqrt{20}/3)^2 + 2 \cdot (4/3)^3 \approx 13.63 \quad (33)$$

Therefore, the maximum value of the function given the set S , is $f(P_1) = f(P_3) = f(P_4) = 16$; and the minimum value is $f(P_2) = -16$.

Problem 1.3

The main Python file used for Problem 1.3. is *genetic_algorithm.py*, which can be found in the folder *Problem_1.3*.

a) For Problem 1.3.a, the Python file *run_single.py* was used, which can be found in the folder *Problem_1.3*.

After exploring different parameters, the original values given in the skeleton file were maintained. Therefore, we chose a tournament size of 2, a tournament probability of 0.75, a crossover probability of 0.8, a mutation probability, p_{mut} of 0.02, and a number of generations of 2000. The original parameters were maintained because, when the parameters were changed, similar results were obtained, even when changing the mutation probability. However, these similarities might be due to the fact that the function under study is "well behaved", facilitating the convergence to the minimum value even with unsuitable parameters.

In this case we perform 10 experiments, and in each of them we calculate the components of the best individual (x_1^*, x_2^*) , which has a fitness value of $1/(g(x_1^*, x_2^*) + 1)$. The Table 2 presents the results obtained.

Experiment	x_1^*	x_2^*	$1/(g(x_1^*, x_2^*) + 1)$	$g(x_1^*, x_2^*)$
1	2.99999768	0.49999939	1.0000	0.0000
2	3.00000751	0.50000207	1.0000	0.0000
3	2.99987787	0.49996959	1.0000	0.0000
4	2.99997205	0.49999224	1.0000	0.0000
5	3.00000095	0.50000028	1.0000	0.0000
6	3.00000215	0.50000058	1.0000	0.0000
7	3.00000751	0.50000207	1.0000	0.0000
8	3.00000751	0.50000207	1.0000	0.0000
9	3.00003105	0.50000922	1.0000	0.0000
10	2.99999052	0.49999700	1.0000	0.0000

Table 2: For each of the 10 experiments performed, the components of the best individual x_1^* and x_2^* are calculated, with $1/(g(x_1^*, x_2^*) + 1)$ being its fitness value (maximum fitness value). In this case, a mutation probability $p_{mut} = 0.02$ was used.

Analyzing the results, the values $(x_1 \approx 3, x_2 \approx 0.5)$ appear to give $g(x_1^*, x_2^*) = 0$, since $1/(g(x_1^*, x_2^*) + 1) \approx 1$. Given that the function $g(x_1, x_2)$ is:

$$g(x_1, x_2) = \underbrace{(1.5 - x_1 + x_1 x_2)^2}_{A(x_1, x_2) \in \mathbb{R}} + \underbrace{(2.25 - x_1 + x_1 x_2^2)^2}_{B(x_1, x_2) \in \mathbb{R}} + \underbrace{(2.625 - x_1 + x_1 x_2^3)^2}_{C(x_1, x_2) \in \mathbb{R}} \quad (34)$$

$$g(x_1, x_2) = A^2(x_1, x_2) + B^2(x_1, x_2) + C^2(x_1, x_2) \geq 0$$

This implies that a point (x_1^*, x_2^*) for which $g(x_1^*, x_2^*) = 0$ could be a potential minimum, since the function is taking its minimum possible value (argument that will be used in Problem 1.3.c), which indicates that the algorithm is working properly.

b) For Problem 1.3.b, the Python file *run_batch.py* was used. This file can be found in the folder *Problem_1.3*.

In this case, we use the same parameters as before, but now we study different mutation probabilities, p_{mut} , and for each, we perform 100 experiments where we calculate the best individual's components (x_1^*, x_2^*) with its fitness value $1/(g(x_1^*, x_2^*) + 1)$. For each value of p_{mut} , we calculate the median of the maximum fitness values obtained for the 100 experiments performed. The Table 3 and the Figure 2 present these results.

The results show how the median of the fitness value converges to 1 when increasing the probability of mutation. As explained above, a fitness value of 1, coincides with $g = 0$, which is the minimum value the function can take, so we are converging to the minimum. In fact, from the mutation probabilities that we have studied, the first p_{mut} that reaches a fitness value of 1 is 0.02, which is $p_{mut} = 1/n$ where n is the number of genes. Therefore, we can argue that $p_{mut} = 1/n$ defines the minimum mutation probability that should be used to reach convergence.

c) In this case, we are interested in estimating which value (x_1^*, x_2^*) returns the actual minimum of the function $g(x_1, x_2)$. That is precisely the reason why we, in the code, maximize the function $(g(x_1, x_2) + 1)^{-1}$.

In Problem 1.3.a, we saw how the best individuals' components $(x_1^* \approx 3, x_2^* \approx 0.5)$ gave values of

p_{mut}	Median
0	0.9902
0.005	0.9977
0.01	0.9993
0.02	1.0000
0.05	1.0000
0.1	1.0000

Table 3: For each value of p_{mut} , which determines the mutation probability, we calculate the maximum fitness value for 100 experiments. With the maximum fitness values obtained from these experiments, we calculate the median. The median for each mutation probability, p_{mut} , is presented in the table.

$g(x_1^*, x_2^*) \approx 0$. As explained above, $g(x_1, x_2)$ can only take values greater than or equal to 0, which explains why we will be taking $(3, 0.5)$ as a guess for the true minimum of the function.

To prove that the function reaches a stationary point at $(x_1^* = 3, x_2^* = 0.5)$, we have to check that it satisfies $\nabla g(x_1, x_2)|_{x_1^*, x_2^*} = 0$.

Given:

$$g(x_1, x_2) = (1.5 - x_1 + x_1 x_2)^2 + (2.25 - x_1 + x_1 x_2^2)^2 + (2.625 - x_1 + x_1 x_2^3)^2 \quad (35)$$

We calculate the gradient:

$$\nabla g(x_1, x_2) = \left(\frac{\partial g}{\partial x_1}, \frac{\partial g}{\partial x_2} \right) \quad (36)$$

$$\begin{aligned} \frac{\partial g}{\partial x_1} &= 2 \cdot (1.5 - x_1 + x_1 x_2) \cdot (-1 + x_2) + 2 \cdot (2.25 - x_1 + x_1 x_2^2) \cdot (-1 + x_2^2) \\ &\quad + 2 \cdot (2.625 - x_1 + x_1 x_2^3) \cdot (-1 + x_2^3) \\ \frac{\partial g}{\partial x_1} \Big|_{x_1=3, x_2=0.5} &= 2 \cdot \underbrace{(1.5 - 3 + 3 \cdot 0.5)}_0 \cdot (-1 + 0.5) + 2 \cdot \underbrace{(2.25 - 3 + 3 \cdot 0.5^2)}_0 \cdot (-1 + 0.5^2) \\ &\quad + 2 \cdot \underbrace{(2.625 - 3 + 3 \cdot 0.5^3)}_0 \cdot (-1 + 0.5^3) = 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial g}{\partial x_2} &= 2 \cdot (1.5 - x_1 + x_1 x_2) \cdot (x_1) + 2 \cdot (2.25 - x_1 + x_1 x_2^2) \cdot (2x_1 x_2) + (2.625 - x_1 + x_1 x_2^3) \cdot (3x_1 x_2^2) \\ \frac{\partial g}{\partial x_2} \Big|_{x_1=3, x_2=0.5} &= 2 \cdot \underbrace{(1.5 - 3 + 3 \cdot 0.5)}_0 \cdot (0.5) + 2 \cdot \underbrace{(2.25 - 3 + 3 \cdot 0.5^2)}_0 \cdot (2 \cdot 3 \cdot 0.5) \\ &\quad + \underbrace{(2.625 - 3 + 3 \cdot 0.5^3)}_0 \cdot (3 \cdot 3 \cdot 0.5^2) = 0 \end{aligned}$$

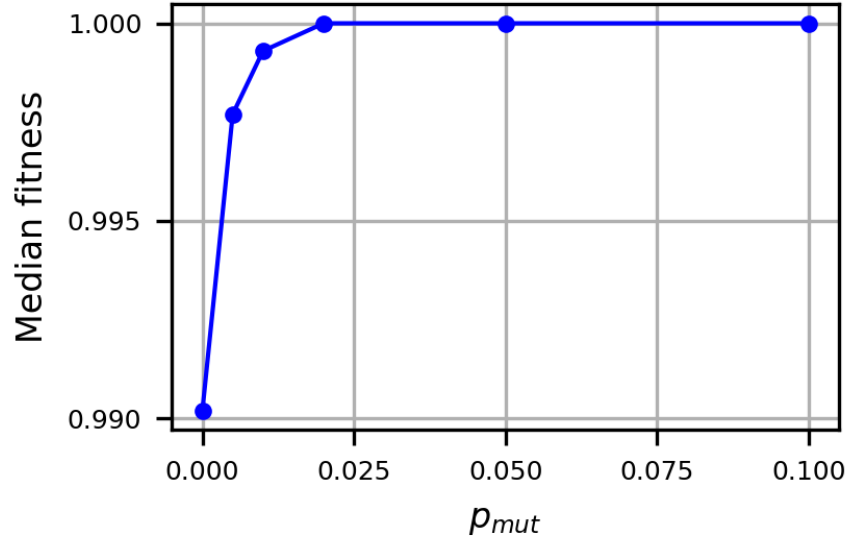


Figure 2: For each value of p_{mut} , which determines the mutation probability, we calculate the maximum fitness value for 100 experiments. With the maximum fitness values obtained from these experiments, we calculate the median. The median for each mutation probability, p_{mut} , is presented in the figure.

Since $\nabla g(x_1, x_2)|_{\mathbf{x}^*} = 0$, we have proved that $\mathbf{x}^* = (3, 0.5)$ defines a stationary point: $[\mathbf{x}^*, g(\mathbf{x}^*)] = [(3, 0.5), 0]$.

Appendix A

In the following table, all the points represented in Figure 1 are presented:

p_{mut}	x_1^*	x_2^*	$f_p(x_1^*, x_2^*)$	$g(x_1^*, x_2^*)$
10^{-4}	0.9992	1.9992	0.0000	3.9952
10^{-3}	0.9922	1.9921	0.0002	3.9529
10^{-2}	0.9329	1.9306	0.0142	3.5973
10^{-1}	0.6963	1.6419	0.3487	2.1808
0.5	0.4972	1.3283	1.1553	1.0114
1	0.4338	1.2102	1.5683	0.6527
3	0.3669	1.0737	2.1167	0.2876
5	0.3480	1.0326	2.2968	0.1874
7	0.3388	1.0122	2.3885	0.1394
10	0.3314	0.9955	2.4650	0.1009
30	0.3186	0.9665	2.6005	0.0356
50	0.3159	0.9602	2.6306	0.0217
75	0.3144	0.9569	2.6461	0.0145
100	0.3137	0.9553	2.6540	0.0109
300	0.3123	0.9519	2.6699	0.0037
500	0.3120	0.9512	2.6731	0.0022
1000	0.3118	0.9507	2.6756	0.0011
1200	0.3118	0.9506	2.6760	0.0009

Table 4: For each value of p_{mut} , the gradient descent method was performed to find the values (x_1^*, x_2^*) for which the function $f_p(x_1, x_2)$ takes its minimum value. The step length used for the gradient descent was 10^{-4} .